

3.6.2 Third-Party Libraries

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں pip install سے انسٹال کیا جاتا ہے۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 23 of 24

3.6.2 Third-Party Libraries

LEARN IT

Third-party libraries are written by the community. They are **not** bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

WORKED EXAMPLE

Installing and importing Pandas (you will use this in Class XII):

```
pip install pandas
import pandas as pd
```

COMMON MISTAKE

`pip` is a **terminal** command, not a Python statement. Do not type `pip install pandas` inside a `.py` file. In VS Code, open the Terminal panel.

اردو میں سمجھیں

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔ نمپائی، پینڈاز، میٹپلاٹ لیب اور ٹینسر فلو اسی قسم کی ہیں۔

CHECK YOURSELF

1. Which library would you choose to draw a bar chart of class marks?
2. Is `math` a third-party library?

Answers: 1. Matplotlib. 2. No — `math` is built-in.

Next lecture: 3.7 Debugging Code in Python